AI.309

Multi-Robot System

COURSE OVERVIEW

Coordinating teams of robots: task allocation, consensus, communication, and collective decision-making.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: TEAMS OF ROBOTS

KEY TOPICS

- Multi-robot vs. swarm
- Coordination challenges
- Centralized vs. decentralized
- Applications

TEACHING FOCUS

Make robots work as a team.

KEY TAKEAWAY Teams outperform individuals on complex tasks.

2

SESSION

60 MINUTES

COMMUNICATION & COORDINATION

KEY TOPICS

- Inter-robot communication
- Consensus algorithms
- Leader election
- Formation control

TEACHING FOCUS

Align robots on shared goals.

KEY TAKEAWAY Coordination requires shared information.

3

SESSION

60 MINUTES

TASK ALLOCATION

KEY TOPICS

- Contract net protocol
- Auction-based allocation
- Optimization methods
- Dynamic repartitioning

TEACHING FOCUS

Assign tasks to the right robots.

KEY TAKEAWAY Efficient allocation maximizes team performance.

4

SESSION

60 MINUTES

COOPERATIVE MANIPULATION

KEY TOPICS

- Joint carrying
- Coordination of multiple arms
- Force synchronization
- Case studies

TEACHING FOCUS

Move large objects together.

KEY TAKEAWAY Cooperation handles heavy or awkward loads.

5

SESSION

60 MINUTES

DECENTRALIZED DECISION MAKING

KEY TOPICS

- Distributed optimization
- Game theory for robots
- Voting and consensus
- Scalable protocols

TEACHING FOCUS

Decide without a central commander.

▲ KEY TAKEAWAY Decentralized teams are resilient.

6

SESSION

60 MINUTES

MULTI-ROBOT SYSTEMS & REVIEW

KEY TOPICS

- Team demo
- Course recap and assessment
- Toward localization (AI.310)
- Team performance eval

TEACHING FOCUS

Evaluate a robot team end-to-end.

▲ **KEY TAKEAWAY** A team is judged by its collective result.